PROJECT REPORT

AI-Based Business Location Selector

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ABSTRACT

The AI-Based Business Location Selector is an educational decision-support prototype designed to help users compare candidate locations for a selected business type and monthly rent budget. The application uses a structured dataset containing attributes such as monthly rent, population, competitor count, daily footfall, accessibility, parking, public transport, income level, and target demand. Machine-learning models—including Linear Regression, Decision Tree, and Random Forest—are trained and evaluated for demand-score prediction. K-Means clustering is used to explore groups of locations, while a suitability-scoring and ranking process orders candidate locations according to illustrative criteria. A Streamlit dashboard presents matching candidates, the highest suitability score, rent information, and the top five recommended locations. Evaluation on the current synthetic dataset produced an R^2 score of 0.9749 for Linear Regression, 0.8417 for Decision Tree, and 0.9380 for Random Forest. These results reflect performance on the generated dataset and do not establish real-world predictive accuracy. The prototype demonstrates how data-driven methods can support preliminary location comparison.

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1. INTRODUCTION

1.1 Background

Selecting a location is an important early decision for a business. Rent, population, customer footfall, competition, accessibility, parking, public transport, and local income can all influence the suitability of a candidate site. Comparing many candidate locations manually can be time-consuming.

1.2 Project Overview

This project provides a simple dashboard where a user can select a business type and a maximum monthly rent budget. The application filters candidate records and displays a suitability overview and a ranked list of recommended locations. Supporting scripts perform model training, clustering, evaluation, and ranking.

1.3 Motivation

The project demonstrates how data preparation, machine learning, clustering, scoring, and visualization can be combined into a single educational decision-support application.

2. PROBLEM STATEMENT, OBJECTIVES AND SCOPE

2.1 Problem Statement

Users may find it difficult to compare multiple possible business locations across several factors at once. The project addresses this by organizing location attributes and presenting filtered, scored, and ranked candidates in a dashboard.

2.2 Objectives

- Prepare and inspect a structured dataset of candidate business locations.
- Train and compare Linear Regression, Decision Tree, and Random Forest models.
- Explore location groups using K-Means clustering.
- Calculate illustrative suitability scores and rank candidate locations.
- Display candidate summaries and the top five locations in a Streamlit dashboard.
- Evaluate model performance and document the limitations of the prototype.

2.3 Scope

The current scope is an academic prototype using synthetic/sample data. It supports preliminary comparisons based on the available fields and illustrative scoring weights. It does not use verified live market, property, demographic, or business-performance data.

2.4 Limitations

- The dataset and demand targets are synthetic/sample values.
- Model scores may reflect patterns created in the sample data rather than real market conditions.
- Suitability scores depend on the implemented scoring rules and selected budget.
- The system should not be used alone to make financial or property-investment decisions.

3. TECHNOLOGIES AND TOOLS USED

- Python: main programming language for data processing and model training.
- Pandas: loading, cleaning, transforming, and exporting tabular data.
- Scikit-learn: machine-learning models, train/test split, metrics, scaling, and K-Means clustering.
- Streamlit: interactive web dashboard.
- Plotly / charting library used by the dashboard: interactive visualizations, where configured.
- Matplotlib: saved clustering visualization, where generated.
- Visual Studio Code: project development environment.
- CSV files: storage format for dataset and generated results.

Note: Confirm the exact libraries in the project's imports and requirements file before final submission. Remove any tool above if it is not used in the final code.

4. DATASET AND METHODOLOGY

4.1 Dataset Description

The project dataset contains 1,000 records and 11 columns. The available fields are: location_name, business_type, monthly_rent, population, competitor_count, daily_footfall, accessibility_score, parking_score, public_transport_score, income_level, and target_demand.

4.2 Data Quality Checks

The recorded evaluation run reported 1,000 rows, 11 columns, no missing values, and no duplicate rows. These checks describe the current dataset file and should be repeated if the data is changed.

4.3 Workflow

1. Load the location dataset.
2. Inspect the columns and basic data quality.
3. Prepare features and target values for supervised model training.
4. Train Linear Regression, Decision Tree, and Random Forest models.
5. Evaluate the models using MAE, RMSE, and R^2.
6. Apply K-Means clustering to explore groups of locations.
7. Calculate suitability scores and rank candidate records.
8. Display budget-matched candidates and the top five ranked locations in Streamlit.

4.4 Suitability Ranking

The ranking component combines location attributes into an illustrative suitability score and sorts candidate records. The dashboard allows users to select a business type and rent budget. Because the score is rule-based and uses synthetic data, its weights and rankings should be interpreted as demonstrative rather than as a validated commercial recommendation.

5. MACHINE LEARNING ALGORITHMS

5.1 Linear Regression

Linear Regression models a target value as a weighted combination of input features. In this project it is evaluated for predicting the target demand value.

5.2 Decision Tree Regressor

A Decision Tree learns a sequence of feature-based splits to predict a numeric target. It can represent non-linear patterns but may overfit if not controlled.

5.3 Random Forest Regressor

Random Forest combines predictions from multiple decision trees. This can improve robustness compared with a single tree, depending on the data.

5.4 K-Means Clustering

K-Means groups records based on feature similarity. The project generated three clusters and recorded a silhouette score of 0.1011. This low score suggests that the clusters are not strongly separated under the current feature representation.

5.5 Evaluation Metrics

MAE (Mean Absolute Error) reports the average absolute difference between predictions and actual target values. RMSE (Root Mean Squared Error) penalizes larger errors more strongly. R^2 indicates how much target variation is explained relative to a baseline on the evaluation split.

6. SYSTEM IMPLEMENTATION

6.1 Main Components

- Dataset: data/business_locations.csv
- Model training: train_model.py
- Clustering: clustering.py
- Location ranking: rank_locations.py
- Model evaluation: evaluate_models.py
- Dashboard: app.py
- Generated results: outputs/ and models/

6.2 Dashboard Features

- Business-type selection.
- Maximum monthly rent budget control.
- Count of matching candidate records.
- Highest suitability score and best candidate rent summary.
- Top five recommended locations table.
- Additional charts and location details, where available in the current dashboard.
- Model prediction or download controls, where available in the current dashboard.

6.3 User Workflow

The user opens the Streamlit application, selects a business type, sets the maximum monthly rent budget, reviews the matching-candidate summary, and examines the ranked top five locations. The displayed recommendations are based on the current dataset and implemented scoring logic.

7. TESTING, EVALUATION AND RESULTS

7.1 Functional Testing

- Dashboard launches successfully.
- Business type can be selected.
- Rent budget control updates the matching candidate list.
- Overview metrics are displayed.
- Top five recommended locations are displayed.
- Generated CSV outputs are available in the outputs folder.

The dashboard functionality above was reported as working during manual testing. Recheck each item after any code or data change.

7.2 Model Evaluation Results

Model	MAE	RMSE	R^2
Linear Regression	78.768	98.637	0.9749
Decision Tree	199.704	247.790	0.8417
Random Forest	123.157	155.044	0.9380

These values are from the previously recorded evaluation run on a 20% test split. They should be reproduced from the current code before final submission. Since the dataset is synthetic, the metrics do not demonstrate equivalent performance on real locations.

7.3 Clustering Result

The current run generated three K-Means clusters with a silhouette score of 0.1011. The score indicates weak cluster separation, so cluster membership should be treated as exploratory.

7.4 Generated Output Files

- outputs/model_comparison.csv — model comparison output.
- outputs/evaluation_results.csv — evaluation metrics.

- outputs/clustered_locations.csv — location records with cluster labels.
- outputs/silhouette_scores.csv — clustering score output.
- outputs/location_clusters.png — clustering visualization.
- outputs/location_ranking.csv — ranked location records.
- outputs/top_5_locations.csv — top five ranked records.

Confirm each filename exists in the current project before describing it as a final deliverable.

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The AI-Based Business Location Selector demonstrates an end-to-end workflow for comparing candidate business locations using tabular data, supervised regression models, clustering, a rule-based suitability score, and a Streamlit dashboard. The application successfully displays budget-matched candidates and a top-five ranked list in the tested interface. The current dataset is synthetic, so the system should be considered an educational prototype rather than a validated commercial decision tool.

8.2 Future Scope

- Replace synthetic data with verified and appropriately licensed real-world location data.
- Validate demand predictions against observed business outcomes.
- Review and justify scoring weights with domain experts.
- Add geographic mapping and verified travel/accessibility information.
- Add model monitoring, input validation, and stronger error handling.
- Test the system on different regions and business categories.

REFERENCES

- Python documentation: https://docs.python.org/3/
- Pandas documentation: https://pandas.pydata.org/docs/
- Scikit-learn documentation: https://scikit-learn.org/stable/
- Streamlit documentation: https://docs.streamlit.io/
- Plotly Python documentation: https://plotly.com/python/

APPENDIX A: PROJECT STRUCTURE

AI-Based Business Location Selector/

- app.py
- train_model.py
- clustering.py
- rank_locations.py
- evaluate_models.py
- data/business_locations.csv
- models/ (trained model files)
- outputs/ (generated evaluation, clustering, and ranking files)

APPENDIX B: BASIC EXECUTION

Run these commands from the project root in the VS Code terminal. Use the sequence that matches the current project scripts and verify that each command completes successfully:

```
python train_model.py

python clustering.py

python rank_locations.py

python evaluate_models.py

python -m streamlit run app.py
```